Zaragoza		Equivariant Systems Theory and Observer Design for Autonomous Systems Robert Mahony, Jochen Trumpf and Tarek Hamel			
M17	22/06/2026	23/06/2026	24/06/2026	25/06/2026	26/06/2026
Time	**Monday**	**Tuesday**	**Wednesday**	**Thursday**	**Friday**
9:00-9:30		**B1: THEORY [RM]**	**C1: THEORY [JT]**	**D1: THEORY [RM]**	**E1: THEORY [RM]**
9:30-10:00		Foundations	Lie Theory and Symmetry	Observer design	Equivariant Systems Theory
10:00-10:30		Symmetry and state		Equivariant Filter	
10:30-11:00		break	break	break	break
11:00-11:30		**B2: CASE STUDY [TH]**	**C2: OPT. CASE STUDY [TH]**	**D2: CASE STUDY [RM]**	**E2: CASE STUDY [TH]**
11:30-12:00		Attitude on SO(3)	Homography on SL(3)	Inertial Navigation	Visual Odometry
12:00-12:30	Registration			System	
12:30-13:00	**A1: INTRO [RM]**				
13:00-13:30	**A2: THEORY [JT]**	LUNCH	LUNCH	LUNCH	LUNCH
13:30-14:00	Matrix Calculus				
14:00-14:30	Matrix ODE	**B3: THEORY [TH]**	**C3: OPT. PRACTICAL [TH]**	**D3: PRACTICAL [JT/TH]**	**E3: OPT. PRACTICAL [TH/JT]**
14:30-15:00	break	Numerical implementation,	observer design	INS filter	Visual Odometry
15:00-15:30	**A3: PRACTICAL [TH]**	bias, time delays	for homography		
15:30-16:00	Matrix Calculus	break	break	break	
16:00-16:30	and examples	**B4: PRACTICAL [JT]**	**C4: THEORY [RM]**	**D4: PRACTICAL [JT/TH]**	
16:30-17:00		Observer design SO(3)	Equivariant Systems:	INS filter	
17:00-17:30	**PYTHON [JT]**	(optional observer SL(3))	kinematics, lifts, extensions, ...		
17:30-18:00	set up computing				